

Customer Behavior Analysis — Task 1 Submission

Alfido Tech Data Analytics Internship Program

Code & Files (GitHub): <https://github.com/ManjuVikram/customer-behavior-analysis>

Executive Summary

Analyzed 250,000 e-commerce transactions from 49,673 customers (Jan 2020–Sep 2023) for Alfido Tech. After cleaning and engineering RFM features, customers were segmented via RFM scoring (cross-validated with K-Means clustering) into 7 groups. Revenue is flat (~\$15M/month, no growth trend) and concentrated: Champions + Loyal Customers are 37.6% of customers but drive 52.5% of revenue. A large Lost segment (18.8% of customers) has gone quiet but still represents a recoverable 8.9% of revenue. The dataset's pre-labeled Churn field was tested and found to carry no behavioral signal (flat ~20% across every segment/cut) — recency-based RFM segment membership is used as the practical churn-risk proxy instead.

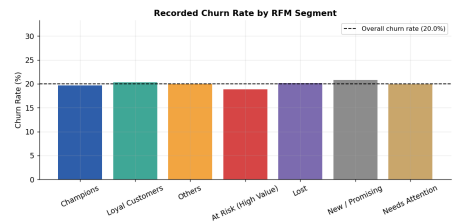
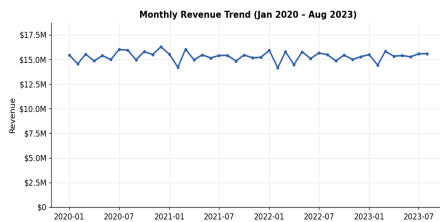
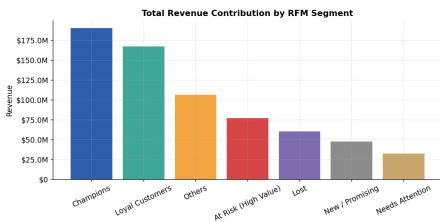
Key Findings

- Revenue flat at ~\$15M/month across 3.5+ years — no growth trend.
- Champions + Loyal (37.6% of customers) generate 52.5% of revenue.
- “Lost” segment (18.8% of customers) inactive ~580 days on avg.
- Clothing/Books lead in orders; Electronics/Home earn more per order.
- Churn label shows no behavioral signal — use recency/RFM instead.

Top Recommendations

- Win back “At Risk (High Value)” and “Lost” segments with targeted offers.
- Build a loyalty/VIP program for Champions & Loyal Customers.
- Convert “New/Promising” buyers to repeat customers via onboarding offers.
- Cross-sell Electronics/Home to frequent Clothing/Books buyers.
- Track a real, time-based churn definition (e.g. 90+ days inactive).

Key Visualizations



Revenue by segment | Monthly revenue trend | Churn rate by segment (flat — key finding)

Notebook Screenshots



Data loading & overview | K-Means clustering (elbow + cluster profile) | Purchase pattern & retention trend charts

Full notebook, executed outputs, and the detailed 9-page analytical report are available in the GitHub repo above.